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Characterization of key odorants causing the roasted and mud-like aromas in strong-aroma types of base Baijiu

Wei Dong[a,b,c], Ruonan Guo[a,b], Miao Liu[d], Caihong Shen[d], Xiaotao Sun[a,b,*], Mouming Zhao[c], Jinyuan Sun[a,b], Hehe Li[a,b], Fuping Zheng[a,b], Mingquan Huang[a,b], Jihong Wu[a,b]

[a] Beijing Advanced Innovation Center for Food Nutrition and Human Health, Beijing Technology and Business University, Beijing 100048, China
[b] School of Food and Chemical Engineering, Beijing Technology and Business University, Beijing 100048, China
[c] College of Food Science and Engineering, South China University of Technology, Guangzhou 510640, China
[d] Luzhou Laojiao Co.Ltd., Luzhou, Sichuan 646000, China

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ABSTRACT

The roasted and mud-like aromas in Chinese strong-aroma types of base Baijiu (base SAB) and the interactions among the corresponding key compounds causing these two characteristic aromas were investigated. A total of 68 and 64 odorants were identified in the base and commercial SAB by comparative aroma extract dilution analysis. Furthermore, the complex matrix of Baijiu always pose a long-term challenge for quantitative accuracy and precision of target aroma compounds; thus, the odorants with flavor dilution (FD) ≥ 9 were further quantified by multiple quantitative techniques, including the direct injection combined with gas chromatography-flame ionization detector, the vortex-assisted surfactant-enhanced emulsification liquid-liquid microextraction (VSLLME) combined with GC-MS, and the derivatization method combined with VSLLME-GC-MS, and 37 and 26 odorants were shown to be important odorants as their OAVs ≥1. Among these components, 5 compounds, hexanoic acid, butyric acid, *p*-cresol, 3-methylindole, and 3-(methylthio)propanal showed higher OAVs in base than in commercial SAB, and based on aroma recombination and omission analysis, they were confirmed to be the key odorants responsible for the mud-like and roasted odors. The evaluation of the aforementioned five key odorants by the "addition" test showed that the roasted and mud-like aromas of base SAB were related, and the odors were mainly attributed to additive or masking effects among the compounds.

1. Introduction

The strong-aroma type of Baijiu (SAB), which is one of the most popular traditionally distilled spirits in China, has gained popularity because of its strong fruity and pineapple-like aromas (Fan & Qian, 2006). Based on the characteristics of Baijiu production, the fresh distillates (base SAB) and the finished product (commercial SAB) are both called SAB. In general, during SAB production, the base SAB is obtained using steam distillation. However, some undesirable aroma characteristics, e.g., "harsh", "green", and "raw", have been detected in base Baijiu (Fan & Qian, 2005). Thus, to balance these aroma characteristics and achieve a consistent quality, commercial SAB is obtained by aging the base SAB, adjusting it to the desired ethanol concentration (38%–52%, *v*/v), and blending. Therefore, it has been suggested that the volatile compounds between base and commercial SAB have different aromatic presentations.

Until now, most studies have been focused on the identification and quantification of volatile compounds in commercial SAB. For instance, in 2005, Fan and Qian first determined 75 volatile compounds in Yanghe Daqu Baijiu by headspace solid-phase microextraction (HS-SPME) coupled with gas chromatography-olfactometry dilution analysis, and the results suggested that esters were the major contributors to its aroma. In addition, 320 compounds were identified in all 18 types of Luzhoulaojiao Baijiu using comprehensive two-dimensional gas chromatography/time-of-flight mass spectrometry (GC × GC/TOF-MS) (Yao et al., 2015). More importantly, most results have demonstrated that fruity, sweet, and floral are the most important odor characteristics in commercial SAB (Fan & Qian, 2006; Xiao, Yu, Niu, Ma, & Zhu, 2016). However, the key aroma compounds in base SAB, as well as the analysis of the aroma difference between base and commercial SAB, have not been elucidated yet.

The aroma of base SAB is generated during the fermentation and distillation process, and is dependent on the peculiar manufacturing process, which can result in base SAB having distinctly different aroma

* Corresponding author at: School of Food and Chemical Engineering, Beijing Technology and Business University, Beijing 100048, China.
E-mail address: sxt_btbu66@163.com (X. Sun).

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Table 1
Concentration of 3-(methylthio)propanal in the other 16 base SAB samples determined using derivatization combined with VSLLME-GC-MS (n = 3).

No.	Sample abbreviation	Cooperant distilleries	Mean concentration	RSD	OAV
			(μg/L)	(%)	
1	LZ–2015	Luzhoulaojiao Distillery Co., Ltd	496.2 ± 40.0	8.1	70
2	LZ–21(14)	Luzhoulaojiao Distillery Co., Ltd	301.5 ± 4.6	1.5	42
3	LZ–21(15)	Luzhoulaojiao Distillery Co., Ltd	717.6 ± 38.8	5.4	101
4	LZ–21(16)	Luzhoulaojiao Distillery Co., Ltd	786.3 ± 49.8	6.3	110
5	LZ–21(17)	Luzhoulaojiao Distillery Co., Ltd	734.2 ± 11.9	1.6	103
6	LZ–21(18)	Luzhoulaojiao Distillery Co., Ltd	659.1 ± 2.7	0.4	93
7	LZ–21(19)	Luzhoulaojiao Distillery Co., Ltd	183.6 ± 6.7	3.6	26
8	LZ–21(20)	Luzhoulaojiao Distillery Co., Ltd	922.3 ± 10.7	1.2	130
9	GJG–(D–9)	Gujinggong Distillery Co., Ltd	626.4 ± 31.8	5.1	88
10	GJG–(D–20)	Gujinggong Distillery Co., Ltd	861.5 ± 15.8	1.8	121
11	GJG–(D–22)	Gujinggong Distillery Co., Ltd	505.3 ± 11.9	2.4	71
12	GJG–(D–24)	Gujinggong Distillery Co., Ltd	700.2 ± 19.7	2.8	98
13	GJG–(D–26)	Gujinggong Distillery Co., Ltd	373.3 ± 23.2	6.2	52
14	GJG–(D–28)	Gujinggong Distillery Co., Ltd	941.1 ± 12.9	1.4	132
15	GJG–(D–30)	Gujinggong Distillery Co., Ltd	780.8 ± 11.2	1.4	110
16	GJG–(D–33)	Gujinggong Distillery Co., Ltd	1552.0 ± 91.2	5.9	218

characteristics compared with the other types of Baijiu (Liu & Sun, 2018; Xiao et al., 2016). Of these steps, fermentation in a unique fermentation container covered with pit mud is the most important process for setting the characteristics of the SAB, as this step endows base SAB with its characteristic mud-like aroma (Jin, Zhu, & Xu, 2017). In our previous work, 3-methylindole, which gives a fecal and animal-like odor, was confirmed to contribute substantially to the off-odor of base SAB (Dong et al., 2018). Moreover, hexanoic acid, butyric acid, and *p*-cresol have also been reported to be involved in this characteristic in commercial SAB (Du, Liu, Wang, & Xu, 2017; Zhao et al., 2018), but little is known about their roles in base SAB. Heat treatment can also be used to change the aroma of Baijiu, especially to achieve a roasted odor. The formation of the corresponding aroma compounds is usually associated with Maillard reaction products generated during the heating of Baijiu (Luo et al., 2018). For instance, according to studies published previously (Zhao, Chen, Li, & Xu, 2018), the process of heat-treating soy sauce-aroma types of Baijiu (SSB) mainly includes making *Daqu* (65–70 °C) and stacking the grains at a high-temperature, as well as fermentation at 42–45 °C and distilling the fermented mixture at 100–105 °C. The characteristic caramel-like, roasted-, and burnt aromas of SSB are formed during its unique manufacturing process. More importantly, based on a reconstitution study, certain volatile sulfide compounds (e.g., dimethyl sulfide, 3-(methylthio)propanal, and 2-furfurylthiol) were determined to be typical potent odorants responsible for the aforementioned aroma in SSB or the roasted sesame-like aroma types of Baijiu (RSB) (Sha, Chen, Qian, Wang, & Xu, 2017; Zheng et al., 2016; M. Zhu, Fan, Xu, & Zhou, 2016). Compared with SSB, the typical process of manufacturing base SAB also involves preparing a medium-temperature *Daqu* (55–60 °C) with fermentation and distillation at relatively high temperatures of 32–35 °C and 95–102 °C, respectively. Moreover, dimethyl sulfide, which is considered the most important odorant, has often been detected in commercial SAB and has an odor activity value (OAV) higher than 1(Wang, Fan, & Xu, 2014). Thus, based on our experience, we hypothesized that the roasted aroma may be generated in base SAB. However, until now, no comprehensive study employing a sensomics approach has been performed on the roasted aroma in base SAB; more importantly, the most important compounds or the individual compounds responsible for specific aroma characteristics remain unclear.

The aroma characteristics of alcoholic beverages are not equal to the sum of all the single odorants present, but instead, the aroma is generated by the various complex interactions among specific aroma compounds (Coetzee et al., 2015; Niu et al., 2018). For instance, Lytra, Tempere, Revel, and Barbe (2012) emphasized that dimethyl sulfide could positively enhance the fruity aroma perception of red wine although it does not possess a fruity aroma itself. Moreover, other interesting findings, including that woody odorants at sub- and perithreshold concentrations could improve the olfactory perception of suprathreshold fruity notes in wine, have been reported (Atanasova et al., 2005). In recent years, the interactions between compounds have been classified into four types, namely, no effects, masking effects, additive action, and synergistic effects (Niu et al., 2018). When studying the interactions among volatile compounds in an oolong tea infusion, (Zhu, Chen, Wang, Niu, & Xiao, 2017) found that mixtures of compounds (such as 3-methylbutanal and 2-methylbutanal) with similar structures or aromas always presented synergistic effects and additive actions, while compounds with dissimilar structures (e.g., 2,4-hexadienal and phenyl acetaldehyde, or acetic acid and hexanoic acid) showed masking effects or weak additive actions. However, until now, the interactions among the aroma compounds in base SAB, especially, the interactions among the key compounds responsible for the mud-like and roasted odors, have not been investigated.

Therefore, the objectives of this study were (i) to clearly identify the differences in the composition of aroma-active compounds between base and commercial SABs using comparative aroma extract dilution analysis in combination with GC-MS/O; (ii) to quantitate the potential important aroma compounds using multiple quantitation methods followed by the calculation of the OAVs; (iii) to clarify the potent odorants responsible for the typical roasted and mud-like odors in base SAB; and (iv) to highlight the interactions between the roasted and mud-like aroma compounds in base SAB.

2. Materials and methods

2.1. Samples of strong-aroma types of Baijiu

Only two types of SAB samples named 'LZ–2016' (64%, alcohol by volume, ABV, base SAB) and 'GJ–1573' (52%, ABV, commercial SAB), were used for the GC-O and GC–MS analyses in this study and they were supplied by Luzhou Laojiao Distillery Co., Ltd. (Sichuan Province, P.R. China). In addition, the data corresponding to the other 16 base SAB samples are presented in Table 1. The representative nature of the samples mentioned above was confirmed by sensory evaluation by three national Baijiu judges. All samples (500 mL from each bottle) were stored at 4 °C until analysis.

2.2. Chemical standards, reagents, and materials

Full details are provided in the Supplementary information.

Table 2
Chemical standards, quantitative ions, and standard curves of the important aroma compounds (FD ≥ 9) in the SAB samples.

No.	Compounds	Methods[a]	Quantified ion (*m/z*)	IS[b]	Linearity				LOD	Recovery
					slope	intercept	Range (mg/L)	R^2	(μg/L)	(%)
Esters										
1	ethyl acetate	DI-FID-DB-WAX	1.637[c]	IS1	0.6371	−0.0126	250–2000	0.9990	45.4	–
2	ethyl isobutyrate	DI-FID-DB-WAX	1.043[c]	IS1	0.9532	−0.0006	2–20	0.9996	25.0	–
3	ethyl butanoate	DI-FID-DB-WAX	1.125[c]	IS1	0.8911	−0.004	120–1200	0.9976	23.0	–
6	ethyl pentanoate	DI-FID-DB-WAX	1.027[c]	IS1	0.9805	−0.0031	50–400	0.9989	217.6	–
8	ethyl hexanoate	DI-FID-DB-WAX	0.974[c]	IS1	1.0249	0.0353	695.2–13,904.0	0.9995	6.3	–
4	ethyl 2-methylbutanoate	VSLLME-HP-5MS-CCl_4	57	IS1	0.2555	−0.0099	0.45–4.5	0.9982	0.1	106.3
5	ethyl isovalerate	VSLLME-HP-5MS-CCl_4	88	IS1	0.2198	−0.0019	1.4–13.5	0.9990	0.2	91.6
7	ethyl 4-methylpentanoate	VSLLME-HP-5MS-CCl_4	89	IS1	0.0426	−0.0022	0.45–4.5	0.9911	4.5	118.9
9	ethyl 3-hexenoate	VSLLME-HP-5MS-CCl_4	69	IS1	0.0303	0.0001	0.1125–1.4	0.9990	0.4	95.3
10	propyl hexanoate	VSLLME-HP-5MS-CCl_4	99	IS1	0.5025	−0.0454	0.1125–30.0	0.9986	0.3	81.5
11	ethyl heptanoate	VSLLME-HP-5MS-CCl_4	88	IS1	0.6098	−0.1711	4.5–75.0	0.9972	0.2	84.8
12	isobutyl hexanoate	VSLLME-HP-5MS-CCl_4	99	IS1	2.1592	−0.0291	0.04–0.8	0.9939	0.6	93.8
13	hexyl butanoate	VSLLME-HP-5MS-CCl_4	71	IS1	0.6984	−0.0027	0.1–10.0	0.9994	0.2	87.8
14	ethyl octanoate	VSLLME-HP-5MS-CCl_4	88	IS1	1.5658	−0.0028	1.4–13.5	0.9978	0.1	89.2
					1.1917	2.2053	22.5–225	0.9963		101.4
15	ethyl 2-hydroxy-4-methyl pentanoate	VSLLME-HP-5MS-CCl_4	69	IS1	0.4993	−0.045	1.4–13.5	0.9938	0.7	105.3
16	hexyl hexanoate	VSLLME-HP-5MS-CCl_4	117	IS1	0.923	−0.0051	0.1125–1.4	0.9968	0.6	101.6
				IS1	0.9429	0.4358	4.5–90	0.9979		80.3
17	ethyl caprate	VSLLME-HP-5MS-CCl_4	88	IS3	0.8427	0.0061	0.009–0.1125	0.9981	0.5	87.5
					1.7724	−0.0791	0.45–4.5	0.9987		88.3
18	3-methylbutyl octanoate	VSLLME-HP-5MS-CCl_4	70	IS1	1.2579	−0.0076	0.1125–1.4	0.9975	0.1	96.3
19	diethyl succinate	VSLLME-HP-5MS-CCl_4	101	IS3	3.1467	−0.3589	0.9–9.0	0.9961	1.1	86.8
20	ethyl tetradecanoate	VSLLME-HP-5MS-CCl_4	88	IS3	0.9697	0.001	0.1125–1.4	0.9927	0.3	113.9
21	ethyl palmitate	VSLLME-HP-5MS-CCl_4	88	IS3	1.41	−0.0966	0.3–3.0	0.9991	0.2	102.0
					1.7058	0.9575	4.5–75.0	0.9938		95.2
Alcohols										
23	3-methyl-1-butanol	DI-FID-DB-WAX	0.764[c]	IS1	1.2203	−0.0229	25–200	0.9982	31.7	–
24	1-butanol	VSLLME-DB-WAX-CH_2Cl_2	56	IS1	0.5754	0.225	45–450	0.9927	6.9	72.0
25	1-hexanol	VSLLME-DB-WAX-CH_2Cl_2	56	IS1	1.0356	0.8363	25–250	0.9944	3.5	99.3
26	1-octanol	VSLLME-DB-WAX-CH_2Cl_2	56	IS1	0.812	−0.0079	0.45–9	0.9995	11.3	86.1
Aromatics										
27	phenylacetaldehyde	VSLLME-DB-WAX-CH_2Cl_2	91	IS1	0.4458	0.077	1.0–10.0	0.9962	19.0	98.6
28	ethyl benzoate	VSLLME-DB-WAX-CH_2Cl_2	105	IS3	1.0761	−0.0059	0.09–4.5	0.9993	1.9	73.3
29	benzene, (2,2-diethoxyethyl)-	VSLLME-DB-WAX-CH_2Cl_2	103	IS3	2.683	−0.0254	0.09–2.3	0.9947	2.2	86.6
30	ethyl phenylacetate	VSLLME-DB-WAX-CH_2Cl_2	104	IS3	1.2177	0.0005	0.09–1.35	0.9962	1.2	78.6
31	phenethyl acetate	VSLLME-DB-WAX-CH_2Cl_2	91	IS3	1.9584	−0.0634	0.9–9.0	0.9932	1.5	114.7
33	ethyl 3-phenylpropionate	VSLLME-DB-WAX-CH_2Cl_2	104	IS3	2.1764	−0.2502	0.45–12.5	0.9993	27.3	120.9
34	phenethyl alcohol	VSLLME-DB-WAX-CH_2Cl_2	91	IS3	2.2903	−0.2536	0.45–7.5	0.9984	2.5	97.0
Phenols										
35	4-methyl guaiacol	VSLLME-DB-WAX-CH_2Cl_2	123	IS3	1.1561	−0.0385	0.25–5.0	0.9982	1.3	111.8
36	4-ethylguaiacol	VSLLME-DB-WAX-CH_2Cl_2	137	IS3	0.7886	0.0041	0.09–1.125	0.9980	3.2	93.6
37	*p*-cresol	VSLLME-DB-WAX-CH_2Cl_2	107	IS3	1.6352	−0.0287	0.45–9.0	0.9974	1.0	118.8
38	4-ethylphenol	VSLLME-DB-WAX-CH_2Cl_2	107	IS3	1.5015	0.0068	0.01–0.5	0.9988	1.9	81.6
39	2-methoxy-4-vinylphenol	VSLLME-DB-WAX-CH_2Cl_2	150	IS3	0.0935	0.0004	0.01–0.45	0.9955	10.3	76.2
41	vanillin	VSLLME-DB-WAX-CH_2Cl_2	151	IS3	0.5837	0.0019	0.01–0.4	0.9981	5.6	81.2
Acids										
42	acetic acid	DI-FID-DB-WAX	1.152[c]	IS2	0.3646	−0.0109	160–1600	0.9943	73.0	–
44	2-methylpropionic acid	DI-FID-DB-WAX	1.493[c]	IS2	0.7312	−0.0016	0.025–2.0	0.9972	3.3	–
48	hexanoic acid	DI-FID-DB-WAX	1.187[c]	IS2	0.9005	−0.0293	200–2000	0.9937	64.2	–
43	propionic acid	VSLLME-DB-WAX-Ether	74	IS2	0.3976	0.0773	1.25–90	0.9993	30.3	80.4
45	butyric acid	VSLLME-DB-WAX-Ether	60	IS2	1.1294	−1.0705	9–450	0.9998	51.7	72.0
46	isovaleric acid	VSLLME-DB-WAX-Ether	74	IS2	1.8894	−0.5391	0.9–45.0	0.9918	28.4	77.0
47	pentanoic acid	VSLLME-DB-WAX-Ether	60	IS2	1.4936	−1.4991	9.0–225.0	0.9998	53.1	86.7
50	4-methylvaleric acid	VSLLME-DB-WAX-Ether	57	IS2	0.6242	−0.0093	0.225–22.5	0.9993	58.3	94.4
51	heptanoic acid	VSLLME-DB-WAX-Ether	60	IS2	1.4034	−1.3183	4.5–112.5	0.9981	26.3	76.4
52	octanoic acid	VSLLME-DB-WAX-Ether	60	IS2	1.285	−1.3519	4.5–112.5	0.9977	62.2	120.7
54	benzeneacetic acid	VSLLME-DB-WAX-Ether	91	IS2	0.0461	−0.0035	0.225–4.5	0.9907	6.5	114.7
55	3-phenyl propionic acid	VSLLME-DB-WAX-Ether	91	IS2	0.2356	−0.009	0.25–5.0	0.9991	88.9	115.2
Sulfur compounds										
56	dimethyl trisulfide	VSLLME-DB-WAX-CH_2Cl_2	126	IS1	0.4851	−0.0007	0.0225–0.9	0.9998	3.6	86.0
57	3-(methylthio)propanal	Derivatization -VSLLME-DB-WAX-CCl4	102	IS5	0.7772	−0.1665	0.1–2.5	0.9970	74.6	82.4
58	3-(methylthio)-propanoic acid ethyl ester	VSLLME-DB-WAX-CH_2Cl_2	74	IS1	0.2542	−0.0003	0.045–0.9	0.9968	4.3	109.3
59	2-thenaldehyde	VSLLME-DB-WAX-CH_2Cl_2	111	IS3	1.1103	−0.0016	0.009–0.9	0.9982	1.1	74.6
60	3-methylthiopropanol	VSLLME-DB-WAX-CH_2Cl_2	106	IS3	0.2755	−0.0011	0.045–0.9	0.9970	2.3	80.0
N-containing compounds										
62	ethyl nicotinoate	VSLLME-HP-5MS-CCl_4	106	IS3	1.0213	−0.0005	0.005–0.9	0.9985	0.1	91.7

(*continued on next page*)

Table 2 (*continued*)

No.	Compounds	Methods[a]	Quantified ion (*m/z*)	IS[b]	Linearity				LOD	Recovery
					slope	intercept	Range (mg/L)	R^2	(μg/L)	(%)
63	3-methylindole	VSLLME-HP-5MS-CCl_4	130	IS4	0.1236	−0.0005	0.009–0.09	0.9976	0.1	116.6
Lactones										
64	γ-nonalactone	VSLLME-DB-WAX-CH_2Cl_2	85	IS3	1.2423	−0.0032	0.045–0.9	0.9986	1.8	119.4
Furans										
67	furfuryl acetate	VSLLME-DB-WAX-CH_2Cl_2	81	IS1	0.1476	0.0001	0.009–0.9	0.9962	4.2	84.1
68	ethyl 2-furoate	VSLLME-DB-WAX-CH_2Cl_2	95	IS1	1.2426	−0.0035	0.025–0.5	0.9982	0.9	115.7

[a] The quantitative methods detailed in the "Multi-Quantitative Analysis of Aroma Compounds" section.
[b] The internal standards used to quantitate the compounds were amyl acetate (IS1), 2-ethylbutyric acid (IS2), ethyl heptadecanoate (IS3), 2,3-dimethylindole (IS4) and 3-(methylthio)butanal-PFBHA (IS5).
[c] Calibration factors of aroma compounds, detected by GC-FID.

2.3. Gas chromatography-mass spectrometry/olfactometry (GC-MS/O) analysis

2.3.1. Isolation of the volatile components

Following a literature procedure with some modifications (Zhao et al., 2018), and associated with some modifications, SAB samples of LZ–2016 and GJ–1573 (50 mL of each) were diluted to 15% ABV with boiled ultrapure water that had cooled to room temperature, and the dilute samples were saturated with NaCl. The diluted Baijiu samples were extracted with CH_2Cl_2 three times (70 mL each time). The combined organic phase I (OPI, 70 mL × 3) was further separated into neutral/basic and acidic fractions (NBF and AF) by the following method.

(a) For the NBF, the aforementioned OPI was extracted three times with sodium carbonate (Na_2CO_3) solution (105 mL each time, pH = 10.0), and then separated to obtain OPII (named as NBF) and the combined aqueous phase (AP).
(b) For the AF, the AP was acidified with 4.0 mol/L HCl solution (pH = 2.0), saturated with NaCl, and extracted 3 times with freshly distilled CH_2Cl_2 (105 mL each time). The organic layers were combined to afford OPIII (AF).

Finally, both fractions were dried by anhydrous Na_2SO_4 overnight, concentrated to 500 μL under a gentle stream of nitrogen, and then prepared for further evaluation by GC–MS/O.

2.3.2. GC-MS/O analysis

The extracts containing the odor-active compounds (NBF and AF) acquired from the 'Isolation of the Volatile Compounds' section were further evaluated by using an olfactory detection port (ODP C200, Gerstel, Germany) coupled to a Trace 1300 GC- ISQ LT GC/MS system (Thermo Fisher Scientific, Waltham, MA, USA). Each sample extract was analyzed on both a DB-WAX column (30 m × 0.25 mm, 0.25 μm film thickness, Agilent Technologies, CA, USA) and an HP-5MS capillary column (30 m × 0.25 mm, 0.25 μm film thickness, Thermo Fisher Scientific, Waltham, MA, USA). The conditions for the GC–MS/O analysis were set according to previous reports (Dong et al., 2018) and are provided in the Supplementary information.

Olfactometry analysis was carried out by 3 trained panelists (including 2 females and 1 male) who are part of the Beijing Key Laboratory of Flavor Chemistry, Beijing Technology and Business University. During the GC analysis, all the panelists independently evaluated the same sample from a sniff port, and then they were asked to record the aroma attributes, retention time (RT), and the intensities of the odor types that were present in the sample extract. To avoid overlooking and mistaking individual odor-active compounds, only the odorants that could be detected by at least two assessors were recorded. Each extract was sniffed in triplicate by every panelist.

2.3.3. Comparative aroma extract dilution analysis (cAEDA)

To further elucidate the contribution of each aroma compounds identified by the aforementioned GC–MS/O method and directly compare the aromas of the LZ–2016 and GJ–1573 samples, cAEDA was applied. The concentrated fractions (NBF and AF) mentioned above were diluted stepwise (1:3, by volume) with CH_2Cl_2, and then evaluated by the GC–MS/O under the same conditions until no odor-active compounds could be identified. The FD factor of each compound was determined as the maximum dilution at which the odor was still detectable. Aroma compounds were identified according to previous reports (Dong et al., 2018) and are detailed in the Supplementary information.

2.4. Multiple quantitative analyses of the aroma compounds

Five methods were used to quantify the aroma compounds (shown in Table 2). Standard calibration curves were used to quantify the target aroma compounds using a suitable capillary column based on the ratio of the peak area of the compound relative to the peak area of the internal standard to determine the concentration of the analyte. The standard calibration curve concentrations and the target compounds were quantified in two SAB model solutions (LZ–2016 base SAB with 64% ABV and a pH of 4.6 and GJ–1573 with 52% ABV and a pH of 3.8). The analysis methods, including recoveries, limits of detection (LODs), and limits of quantification (LOQs), are detailed in the Supplementary information.

2.4.1. Direct injection combined with GC-FID analysis

The major compounds, including ethyl acetate, ethyl isobutyrate, ethyl butanoate, ethyl pentanoate, ethyl hexanoate, 3-methyl-1-butanol, acetic acid, 2-methylpropionic acid and hexanoic acid, were directly quantified by GC-FID equipped with a DB-WAX column (30 m × 0.25 mm, 0.25 μm film thickness). Amyl acetate (IS1, 11,630 mg/L) was used as the internal standard. Samples of the two types of SAB (1 mL) were spiked with 20 μL of IS1 to a final concentration of 232.6 mg/L. Afterwards, 1 μL of each test sample was injected in split mode (25:1). Nitrogen at a constant flow rate of 1 mL/min was used as the carrier gas. The oven temperature program followed the same steps as were used for GC analysis described in the 'GC–MS/O Analysis' section. The injector and FID temperatures were both set at 250 °C.

2.4.2. The vortex-assisted surfactant-enhanced emulsification liquid-liquid microextraction (VSLLME) combined with GC-MS analysis

According to the method published previously by our group (Dong et al., 2018), minor compounds with the exception of 3-(methylthio) propanal were quantitated using VSLLME-GC–MS. A diluted Baijiu sample (15% ABV, 5.0 mL) with 50.0 μL of the mixed internal standard solution (detailed in Table 2) was transferred to a 10.0-mL glass vial

Table 3
Concentrations and Odor Activity Values (OAVs) of the SAB samples.

No.	Compounds	Odor threshold	Concentrations (μg/L)		OAV	
			av ± SD (n = 3)			
		(μg/L)	LZ–2016 base SAB	GJ–1573	LZ–2016 base SAB	GJ–1573
8	ethyl hexanoate	55.3[c]	7,606,936 ± 17,161	1,487,026 ± 6408	137,558	26,890
3	ethyl butanoate	81.5[c]	622,200 ± 1786	217,932 ± 369	7634	2674
6	ethyl pentanoate	26.8[c]	143,889 ± 1500	44,849 ± 2636	5369	1673
14	ethyl octanoate	12.9[d]	43,878 ± 1045	2382 ± 17	3401	185
56	dimethyl trisulfide	0.4[c]	187 ± 12	26 ± 2	467	65
48	hexanoic acid	2520.0[c]	1,126,596 ± 34,264	722,576 ± 17,196	447	287
45	butyric acid	964.0[c]	295,816 ± 40,762	78,618 ± 8931	307	82
5	ethyl isovalerate	7.0[c]	1847 ± 4	2783 ± 60	264	398
2	ethyl isobutyrate	57.0[c]	9437 ± 1292	8766 ± 304	166	154
7	ethyl 4-methylpentanoate	6.0[a]	932 ± 20	605 ± 2	155	101
47	pentanoic acid	389.0[c]	56,841 ± 8653	13,732 ± 1660	146	35
1	ethyl acetate	32,600.0[c]	1,772,911 ± 17,586	985,274 ± 4023	54	30
23	1-butanol	2730.0[c]	147,160 ± 8188	41,802 ± 2478	54	15
57	3-(methylthio)propanal	7.12[d]	369 ± 31	–	52	–
4	ethyl 2-methylbutanoate	24.7[a]	863 ± 21	819 ± 30	35	33
37	*p*-cresol	167.0[c]	3271 ± 201	571 ± 19	20	3
25	1-hexanol	5370.0[d]	97,718 ± 5482	30,025 ± 1548	18	6
16	hexyl hexanoate	1890.0[e]	28,416 ± 794	146 ± 4	15	< 1
33	ethyl 3-phenylpropionate	125.0[c]	1646 ± 88	635 ± 20	13	5
46	isovaleric acid	1050.0[c]	13,702 ± 2649	3885 ± 497	13	4
44	2-methylpropionic acid	1580.0[d]	1130 ± 167	95 ± 12	11	–
52	octanoic acid	2700.0[e]	25,675 ± 4066	5535 ± 501	10	2
50	4-methylvaleric acid	144.0[e]	1311 ± 218	300 ± 62	9	2
63	3-methylindole	6.09[b]	46 ± 3	–	8	–
27	phenylacetaldehyde	262.0[d]	1903 ± 207	1498 ± 84	7	6
42	acetic acid	160,000.0[d]	732,061 ± 106,844	342,535 ± 236,447	5	2
31	phenethyl acetate	407.0[c]	2075 ± 112	992 ± 45	5	2
11	ethyl heptanoate	13,200.0[c]	44,450 ± 1143	5859 ± 91	3	< 1
26	1-octanol	1100.0[e]	2101 ± 101	439 ± 28	2	< 1
51	heptanoic acid	13,300.0[e]	23,926 ± 3848	5382 ± 633	2	< 1
36	4-ethylguaiacol	123.0[c]	130 ± 8	–	1	–
35	4-methyl guaiacol	315.0[d]	425 ± 23	–	1	–
64	γ-nonalactone	91.0[d]	129 ± 14	59 ± 3	1	1
55	3-phenyl propionic acid	420.0[a]	374 ± 38	473 ± 16	1	1
43	propionic acid	18,100.0[c]	13,666 ± 156	1935 ± 63	1	< 1
17	ethyl caprate	1120.0	839 ± 22	13 ± 2	1	< 1
10	propyl hexanoate	12,800.0[c]	28,416 ± 794	146 ± 4	1	< 1
24	3-methyl-1-butanol	179,000.0[c]	84,178 ± 144	142,068 ± 324	< 1	1
28	ethyl benzoate	1430.0[c]	174 ± 9	62 ± 2	< 1	< 1
30	ethyl phenylacetate	909.0[c]	341 ± 23	61 ± 4	< 1	< 1
38	4-ethylphenol	618.0[c]	11 ± 1	49 ± 1	< 1	< 1
12	isobutyl hexanoate	5250.0[e]	461 ± 2	81 ± 1	< 1	< 1
13	hexyl butanoate	8620.0[f]	3483 ± 71	232 ± 4	< 1	< 1
67	furfuryl acetate	26,050.0[a]	114 ± 9	13 ± 1	< 1	< 1
68	ethyl 2-furoate	132,000.0[f]	31 ± 1	166 ± 9	< 1	< 1
19	diethyl succinate	353,193.0[a]	1276 ± 91	1715 ± 76	< 1	< 1
34	phenethyl alcohol	28,900.0[c]	1808 ± 92	1342 ± 49	< 1	< 1
54	benzeneacetic acid	1430.0[f]	469 ± 74	415 ± 75	< 1	< 1
41	vanillin	439.0[c]	58 ± 3	11 ± 1	< 1	< 1
58	3-(methylthio)-propanoic acid ethyl ester	3080.0[f]	308 ± 16	–	< 1	–
60	3-(methylthio)propanol	2110.0[c]	205 ± 12	–	< 1	–
39	2-methoxy-4-vinylphenol	209.3[f]	12 ± 0.2	–	< 1	–
9	ethyl 3-hexenoate	–	473 ± 20	112 ± 6	–	–
18	3-methylbutyl octanoate	–	141 ± 1	–	–	–
15	ethyl 2-hydroxy-4-methyl pentanoate	–	4268 ± 32	6739 ± 115	–	–
59	2-thenaldehyde	–	22 ± 1	11 ± 0.4	–	–
29	benzene, (2,2-diethoxyethyl)-	–	1575 ± 17	163 ± 1	–	–
62	ethyl nicotinoate	–	15 ± 1	9 ± 1	–	–
20	ethyl tetradecanoate	–	731 ± 7	300 ± 0.2	–	–
21	ethyl palmitate	–	14,044 ± 445	601 ± 9	–	–

[a] Odor thresholds in 46% ethanol-water solution detected in this work.
[b] Odor threshold reported in (Dong et al., 2018).
[c] Odor threshold reported in (Fan & Xu, 2011).
[d] Odor threshold reported in (Gao, Fan, & Xu, 2014).
[e] Odor threshold reported in (Zhao et al., 2018).
[f] Odor threshold reported in (Sun et al., 2018).

with a conical bottom and then saturated with NaCl (1.2 g). Based on the complex chemical structures and the difference in the polarities of the target aroma-active compounds, various combinations of surfactants (as emulsifiers) and extraction solvents were optimized and used for compound extraction. The separations were finally achieved on either DB-WAX or HP-5MS column:

(a) *Quantitation of the N-containing compounds, including 3-methylindole and ethyl nicotinate, and the esters:* Tween 20 (250 μL, 1.0 mmol/L) with 500 μL of CCl_4 allowed sufficient separation on the HP-5MS column.
(b) *Quantitation of the highly polar acids:* CTAB (250 μL, 1.0 mmol/L) with 500 μL of ether allowed sufficient separation on the DB-WAX column.
(c) *Quantitation of the other aroma-active compounds:* CTAB (250 μL, 1.0 mmol/L) with 500 μL of CH_2Cl_2 allowed sufficient separation on the DB-WAX column.

In addition, as part of every extraction step (a–c), the mixture was shaken vigorously on an IKA VORTEX 2 vortex agitator (Janke & Kunkel, Staufen, Germany) at 701.2 ×*g* for 30 s and centrifuged at 3778.8 ×*g* for 10 min at room temperature, leaving the final extract at the bottom of the vial, and each sample was analyzed by injection into the GC–MS system. The GC–MS conditions were the same as those described in the 'Gas Chromatography-Mass Spectrometry/Olfactometry (GC-MS/O) Analysis' section. The SAB samples were analyzed in triplicate.

2.4.3. Derivatization combined with VSLLME-GC-MS in selective ion monitoring (SIM) mode

In this study, modified from literature (Rodríguez-Bencomo, Schneider, Lepoutre, & Rigou, 2009), 3-(methylthio)propanal was quantified after derivatization with *O*-(2,3,4,5,6-pentafluorobenzyl) hydroxylamine hydrochloride (PFBHA) using GC–MS with a DB-WAX column. Five milliliters of the dilute Baijiu sample (15% ABV) was transferred into a 10.0-mL glass vial with a conical bottom and saturated with NaCl (1.2 g). After 200 μL of 3-(methylthio)butanal (IS5, 5.0 mg/L in ethanol) and 50 μL of PFBHA (20.0 g/L in ethanol) were added, the tube was immediately capped, and the mixture was shaken vigorously on an IKA VORTEX 2 vortex agitator. The sample was left to derivatize at 45 °C for 30 min, then extracted with 500 μL of CCl_4 under shaking on an IKA VORTEX 2 vortex agitator at 701.2 ×g for 30 s, and then centrifuged at 3778.8 ×*g* for 10 min at room temperature. The final extract was left at the bottom of the tube and injected into the GC–MS system.

During the GC analysis (in split mode, 10:1), a constant flow rate (1 mL/min) of helium (the carrier gas) was maintained, and the temperature of the injector was held at 250 °C. The oven temperature was initially held at 40 °C, raised to 205 °C at a rate of 10 °C/min, held for 5 min, raised to 245 °C at 15 °C/min, and finally held for 7 min. The mass spectrometer (*m/z* range of 45–300) was operated in electron ionization mode at 70 eV with SIM. The selected ions of IS5 after derivatization were *m/z* 61, 75, and 116. The quantitation ions chosen for 3-(methylthio)propanal after derivatization were *m/z* 61, 75, and 102.

2.5. Sensory evaluation

2.5.1. Trained assessors and general conditions

Ten independent panels of trained assessors (five females and five males, average age of 25 years old, recruited from the Key Laboratory of Flavor Chemistry at Beijing Technology and Business University) participated in the sensory evaluation. They were selected from 25 candidates on the basis of the guidelines provided by the ISO 8586:2012 standard (ISO, 2012) and all panelists had > 2 years of experience in fields related to Baijiu. Then, basic training, including operation procedures and terms of sensory evaluation, the typical flavor characteristics of different Baijiu and the discrimination of Baijiu qualities and types, was carried out by national Chinese Baijiu judges from Luzhou Laojiao Co. Ltd. (China).

During the whole sensory analyses, samples were evaluated in standard sensory booths (ISO, 2007), with white lighting, controlled airflow, and a room temperature of 20–25 °C.

2.5.2. Experimental procedure

2.5.2.1. Descriptive profile analysis. For the aroma profile analysis, two specific training sessions were performed similarly to a previously published method (Ares et al., 2018; Fan & Qian, 2005). In the first phase, 68 aroma standards, shown in Table 3, were prepared in concentrations 100-fold above their recognition thresholds in water and used to help the assessors describe the odor qualities of each odorant. After a 3-month training and testing period, a 10-member-well-trained panel (five females and five males, average age of 25 years old) underwent an additional 6 h of training to become familiar with the evaluation of two types of tested SAB. Two samples (30 mL each), were each placed in a separate low-density polyethylene wash bottle (250 mL) with the siphon tubes removed from the caps, equilibrated at room temperature for 20 min, and then assessed by the aforementioned panelists. During the descriptive analysis, panelists were asked to evaluate aroma compositions of each sample by gently squeezing the bottle and taking short sniffs of the air emitted from the nozzle until the panelists reached a consensus on the organoleptic terms of each SAB sample. After this step, eight aroma descriptors, namely, fruity, floral, sweet, acidic/rancid, alcoholic, roasted, green, and mud-like, were selected. In the second phase, the intensities of the individual characteristics and the overall aroma qualities based on the aforementioned eight aromatic attributes were ranked on an 11-point scale from 0 (not perceivable) to 5.0 (strongly perceivable). Finally, each term was anchored with a standard reference with an intensity rating determined by the aforementioned panel, as follows: "fruity" represented by 5530.0 mg/L of ethyl hexanoate (intensity score of 4.0); "floral" represented by 2890.0 mg/L of phenethyl alcohol (intensity score of 5.0); "sweet" represented by 9.1 mg/L of γ-nonalactone (intensity score of 4.5); "acidic/rancid" represented by a mixture of 16,000.0 mg/L acetic acid and 96.4 mg/L butyric acid (intensity score of 4.0); "alcoholic" represented by 65% ABV ethanol-water solution (15 mL, intensity score of 4.0); "roasted" represented by 73.0 mg/L of 2,3,5-trimethylpyrazine (intensity score of 4.0); "green" represented by 110.0 mg/L of 1-octanol (intensity of 5.0); and "mud-like" represented by 2.0 g of pit mud (intensity score of 4.0). The judgments of the individual panelists were averaged. Each sample was analyzed in triplicate.

2.5.2.2. Aroma recombination of two types of SAB (LZ–2016 base SAB and GJ–1573). The aroma compounds with OAVs ≥1 in the two types of SAB were combined according to their detected concentrations to prepare two stock model solutions (LZ–2016 with 64% ABV and pH = 4.6, and GJ–1573 with 52% ABV and pH = 3.8) (Table 3). The mixed stock solutions and samples of the respective SABs were equilibrated for at least 20 min at room temperature, prior to the descriptive analysis, as described in the "Descriptive profile analysis" section.

2.5.2.3. Aroma omission test by discrimination analysis. A triangle test was carried out to determine the significance of each odorant to the overall aroma of the base SAB sample. In this step, 13 aroma omission models were prepared by omitting one or a group of selected compounds from the mixed solution used to model the aroma of the LZ–2016 base SAB sample. Each omission model (15 mL) was evaluated against two complete models' solutions prepared by mixing the standard aroma compounds at their concentrations in the LZ–2016 base SAB sample (Table 3), then the aforementioned three samples, which were identified using randomly assigned three-digit codes, were

sniffed by 10 trained panelists to identify the solution that was different. Each test was repeated in triplicate, and the significance of the difference between the omission model and the complete odor model solution was evaluated according to the method described in the "Statistical analysis" section.

2.5.2.4. Analysis of the interactions. As has been described by numerous researchers, OAVs, or concentration/threshold ratios of mixtures of odorants at sub-threshold values, are approximately additive. In other words, if one mixture contains *n* odorants at levels below threshold, and the summation of the *n* concentration/threshold ratios is *y*, then that mixture is *y* times above threshold (Culleré, Cacho, & Ferreira, 2007; Ferreira, 2012). To evaluate the interactions among compounds, in our study, the mixture containing the test compounds was added to the stock model solutions (LZ–2016 with 64% ABV and pH = 4.6) according to the concentrations of the components in the real tested base SAB samples. Then, the mixture was diluted stepwise until the threshold was reached, and the final dilution times were recorded as OAV_{mix}. Finally, each OAV_{mix} obtained experimentally from the mixture was further compared with a simple additive theory model (ΣOAV_i) using the following equation:

$$\sum_i^n OAV = \mathrm{OAV1} + \mathrm{OAV2} + \cdots + \mathrm{OAV}n\mathrm{R} = \sum_i^n OAV/\mathrm{OAVmix}$$

All the experiments were performed in triplicate. According to this method, the interactions between compounds can be categorized into four types: (i) no effect ($R = 1$), which is when the sum of the individual OAVs of the compounds equals the OAV of their mixture; (ii) masking effect ($R > 1$), which is when the sum of the individual OAVs was higher than the OAV for the mixture; (iii) additive action ($0.5 < R < 1$) and (iv) synergistic effects ($R < 0.5$) are both associated with the sum of the OAVs of the individual compounds being less than the OAV of their mixture (Ferreira, 2012; Guadagni, Miers, & Venstrom, 1969; Zhu et al., 2017).

2.6. Statistical analysis

Details of the calculations used for statistical analysis are provided in the Supplementary information.

3. Results and discussion

To initially determine the differences in the overall aromas of LZ–2016 base SAB and GJ–1573, aroma profile analyses were performed as described in the "Descriptive Profile Analysis" section. As shown in Fig. 1, both samples showed similar intensities of fruity, floral, sweet, acidic/rancid and green attributes; however, the intensities of their alcoholic, roasted and mud-like odors were significantly different ($p < 0.01$). More importantly, these were the dominate odors in LZ-2016 base SAB. Notably, the roasted and mud-like aroma attributes were identified or comprehensively identified in base SAB for the first time herein. Therefore, to further analyze the odorants responsible for the aroma differences between the two tested SAB samples, cAEDA was applied.

3.1. Identification of the aroma compounds by GC-MS/O analysis of two types of SAB

cAEDA on the NBF and AF from each tested SAB samples obtained by LLE found 68 odorants with an FD factor in range of 1–59,049 in the LZ–2016 base SAB sample. Furthermore, 64 odor-active components were determined in GJ–1573, and of these, 64 were common to both SAB samples. The flavor dilution chromatograms obtained from cAEDA of the two SABs are shown in Fig. 2.

Among these compounds, ethyl hexanoate, which has a fruity odor, ethyl 3-phenylpropionate, which has a rose and honey odor, butyric

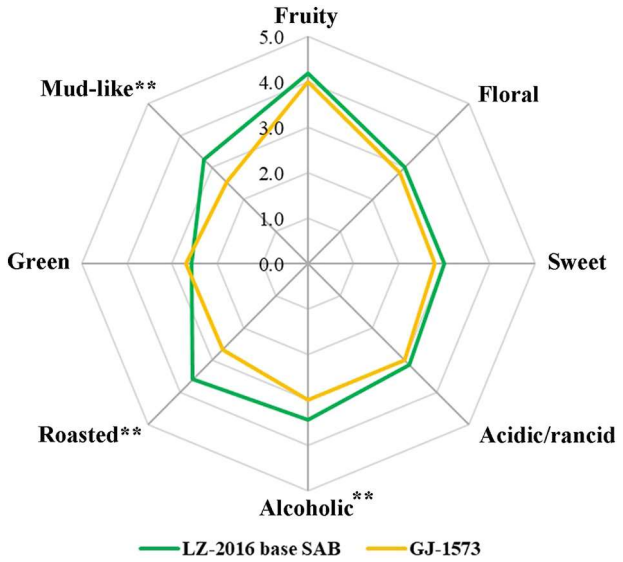


Fig. 1. Aroma profiles of base (green) and commercial SAB (yellow). Significance is indicated at ** $p < 0.01$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

acid and isovaleric acid, which have an cheesy/sweaty odors, gave the highest FD factors of 59,049. These are commonly reported as the characteristic aroma compounds in SAB, regardless of its origin (Wang et al., 2014). In addition, ethyl butanoate, ethyl 2-methylbutanoate, ethyl isovalerate, ethyl pentanoate, ethyl 4-methylpentanoate, diethyl succinate, *p*-cresol, and hexanoic acid, which showed relatively high FD values ($6561 \leq FD \leq 19{,}683$), might also be major contributors to the characteristic aromas of the two SAB samples, giving them fruity, sweet, animal-like, and rancid aromas.

Although most aroma-active compounds were common to both SABs, the FD values were considerably different (detailed in the Supplementary information, Table S1). The majority of the odorants (47 odorants) were found to be higher in the LZ–2016 base SAB than in GJ–1573. The most significant differences were found for 3-(methylthio)propanal, 3-methylthiopropanol, and 3-(methylthio)-propanoic acid ethyl ester. These roasted potato and onion-like odorants showed high FD factors (27–729) in LZ–2016 but were not sensorially detectable in GJ–1573, which might explain why a more intense roasted odor was noted in LZ–2016 than in GJ–1573 in the sensory evaluation (Fig. 1). More importantly, this is the first time that the aforementioned three sulfur compounds have been detected and rated as important aroma compounds in base SAB samples. Among these compounds, 3-(methylthio)propanal, which contributed roasted potato flavors, had the highest FD value (729). According to the literature described by Escudero, Hernández-Orte, Cacho, and Ferreira (2000), Strecker degradation of methionine was the most likely pathway for the formation of 3-(methylthio)propanal. Moreover, 3-(methylthio)propanal could also be formed by sulfur metabolism by *Saccharomyces cerevisiae* yeast (Ugliano & Henschke, 2009). This compound has been widely detected in red wine, Chinese Vidal ice wine, RSB, and other beverages. Furthermore, it was determined to be the primary odorant responsible for the roasted aroma of RSB (Sha et al., 2017; Zheng et al., 2016). However, the contribution of 3-(methylthio)propanal to the aroma profile of base SAB was not clear, and recombination and omission tests were therefore conducted. Additionally, 3-methylindole, which was newly confirmed by our group to be the main contributor responsible for the mud-like off-odor in base SAB, is worth mentioning (Dong et al., 2018).

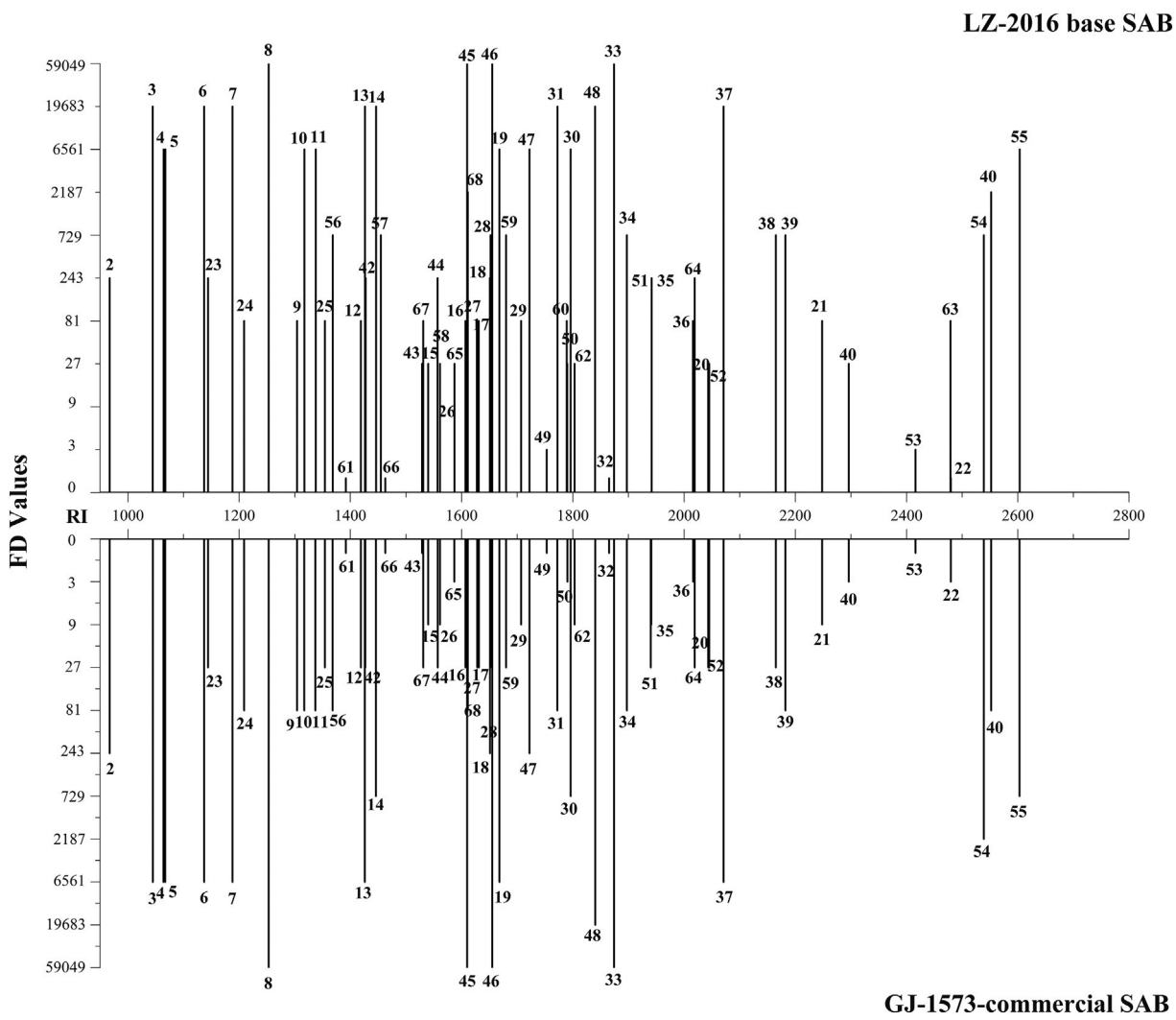


Fig. 2. Comparison of the FD values of the aroma-active compounds in LZ–2016 base SAB and GJ–1573. The labels of the aroma compounds correspond to those listed in Table S1.

3.2. Quantitative analysis of the selected aroma compounds and their odor activity values

3.2.1. Method development

The data obtained from the cAEDA are helpful for screening aroma-active compounds from the various odorless volatile compounds present. However, to further evaluate the influence of the matrix of the tested SAB samples on the overall aroma impression, the collection of quantitative data and the determination of the OAVs are also necessary. The compounds for quantitation were selected based on their high FD values (FD ≥ 9) in the AEDA, and 62 and 53 odorants in LZ–2016 and GJ–1573, respectively, were quantitated using the various methods mentioned in the "Multiple Quantitative Analyses of the Aroma Compounds" section. The employed quantitative methods were able to determine all the aroma-active compounds in the tested SAB samples; moreover, as shown in Table 2, good linearities were found with correlation coefficients (R^2 values) ranging from 0.9907 to 0.9998 and RSDs ≤10% (n = 3). However, the method for the quantitation of 3-(methylthio)propanal was derivatized with PFBHA as shown in Fig. 3A, and the structure of the derivatized product was confirmed by MS analysis (Fig. 3B). The mass spectrum showed a signal at m/z 299 corresponded to the [M^+] ion of 3-(methylthio)propanal after derivatization; moreover, the incorporation of the amine group (-NH_2-) [m/z 102 (Fig. 3B)] of PFBHA replaces the oxygen atom on the carbonyl of 3-(methylthio)propanal with a nitrogen atom, resulting in a specific fragment transition from m/z 104 to m/z 102. Moreover, the m/z 75 and 61 fragments are attributed to the losses of -C_3H_7S and -C_2H_5S, respectively. Finally, we choose the m/z 102 ion as the quantification ion and the other two ions (m/z 75 and 61) as qualifiers. As shown in Fig. 3C, the 3-(methylthio)propanal peak after derivatization was enhanced by factor of > 100 compared to the nonderivatized peaks, which allow the facile identification of 3-(methylthio)propanal separate from other analytes in the matrix. For the quantitation of 3-(methylthio)propanal, the internal standard, 3-(methylthio)butanal (IS5), was also derivatized by PFBHA, and the selected ions were m/z 116, 75, and 61.

3.2.2. Odorant concentrations

As shown in Table 3, ethyl hexanoate was present in the highest concentrations (1487.0–7606.9 mg/L), followed by ethyl butanoate, hexanoic acid, ethyl acetate, and acetic acid. These 5 compounds were all present at levels above 100 mg/L, and they were taken as the key volatile components of SAB. Additionally, ethyl pentanoate (44.8–143.9 mg/L), pentanoic acid (13.7–56.8 mg/L), 1-butanol (41.8–147.2 mg/L), 1-hexanol (30.0–97.7 mg/L) and 3-methyl-1-butanol (84.2–142.1 mg/L) were also present in high concentrations (≥ 10 mg/L). As expected, ethyl octanoate (2.4–43.9 mg/L), isovaleric acid (3.9–13.7 mg/L), *p*-cresol (0.6–3.3 mg/L), ethyl 3-phenylpropionate

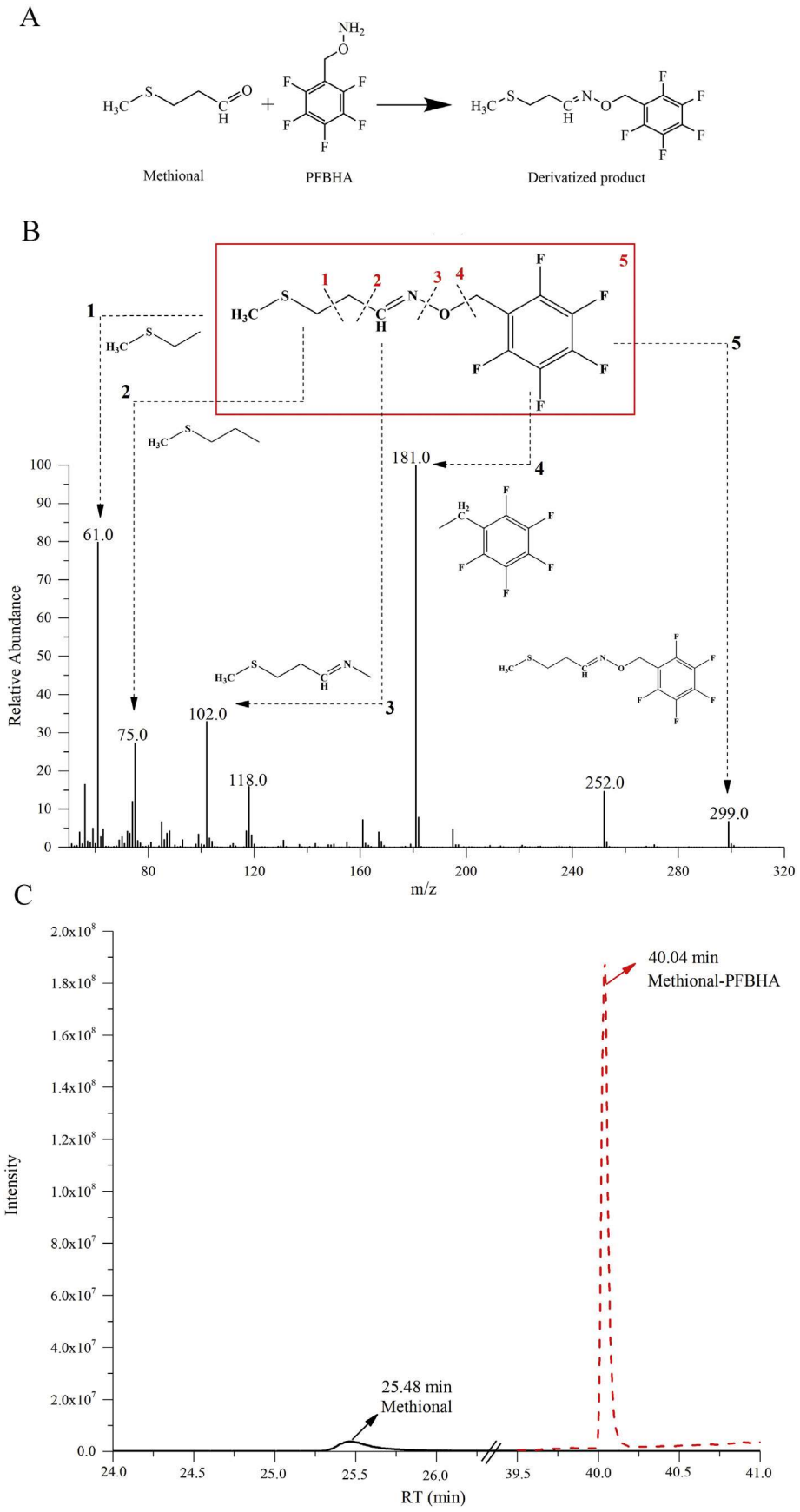


Fig. 3. A. Derivatization scheme showing the reaction of 3-(methylthio)propanal with PFBHA; B. Structures of derivatized 3-(methylthio)propanal showing the fragments used for quantitation. C. Typical chromatogram obtained in SIM mode: black represents the nonderivatized peak of 3-(methylthio)propanal; red represents the intensity of the peak obtained by derivatization of 3-(methylthio)propanal. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4
Comparison of important odorants in LZ–2016 base SAB and GJ–1573.

NO.	Compounds	FD factor (FD ≥ 9)		OAV		OAV ratio[a]
		LZ-2016 base SAB	GJ-1573	LZ-2016 base SAB	GJ-1573	LZ-2016 base SAB/GJ-1573
16	hexyl hexanoate	81	27	15	< 1	195
10	propyl hexanoate	6561	81	1	< 1	195
17	ethyl caprate	81	27	1	< 1	65
14	ethyl octanoate	19,683	729	3401	185	18
44	2-methylpropionic acid	243	27	11	< 1	12
11	ethyl heptanoate	6561	81	3	< 1	8
56	dimethyl trisulfide	729	81	467	65	7
37	*p*-cresol	19,683	6561	20	3	7
26	1-octanol	27	9	2	< 1	5
45	butyric acid	59,049	19,683	307	82	4
47	pentanoic acid	6561	243	146	35	4
63	3-methylindole	81	–	8	–	–
57	3-(methylthio)propanal	729	–	52	–	–
36	4-ethylguaiacol	81	–	1	–	–
35	4-methyl guaiacol	243	–	1	–	–
58	3-(methylthio)-propanoic acid ethyl ester	27	–	< 1	–	–
60	3-methylthiopropanol	81	–	< 1	–	–
39	2-methoxy-4-vinylphenol	729	–	< 1	–	–

[a] OAV ratio: The OAV ratios of 18 aroma compounds between LZ–2016 base SAB and GJ–1573.

(0.6–1.6 mg/L), γ-nonalactone (59.0–129.0 μg/L) and dimethyl trisulfide (26.0–187.0 μg/L), which have frequently been described as important aroma compounds in SAB (Wang et al., 2014; Zhao et al., 2018), also appeared in high concentrations.

3.2.3. Odor activity values (OAVs)

37 and 26 odorants were further verified as important aroma compounds in the tested SAB samples with OAVs ≥1 (Table 3). In LZ–2016 base SAB, ethyl hexanoate was calculated to have the highest OAV (137558), followed by ethyl butanoate (OAV 7634), ethyl pentanoate (OAV 5369), ethyl octanoate (OAV 3401), and dimethyl trisulfide (OAV 467); however, the highest OAVs among the 26 odorants in GJ–1573 were found for ethyl hexanoate (OAV 137558), ethyl butanoate (OAV 2674), ethyl pentanoate (OAV 1673), ethyl isovalerate (OAV 398), and hexanoic acid (OAV 287).

3.2.4. Comparison of important odorants in LZ–2016 base SAB and GJ–1573

As shown in Table S1, 42 odorants, that with FD ≥ 9 in both samples, were found to have different FD values in LZ–2016 base SAB and in GJ–1573. Moreover, considering their respective odor thresholds in Baijiu, the main differences between the aforementioned two samples can be further attributed to 18 compounds, of which OAVs were increased in LZ–2016 base SAB at least by a factor of 4 (Matheis & Granvogl, 2016). As shown in Table 4, the OAV ratios of the 18 odorants between LZ–2016 base SAB and in GJ–1573 were in a range of 4–195.

Among these 18 compounds, 3-(methylthio)propanoic acid ethyl ester, 3-(methylthio)propanol, and 2-methoxy-4-vinylphenol were only detected in LZ–2016 with OAVs of < 1 and therefore did not contribute to the overall aroma. Of the 15-remaining active-aroma compounds, 5 odorants (ethyl octanoate, dimethyl trisulfide, *p*-cresol, butyric acid, and pentanoic acid) showed OAVs ≥1 in both tested SAB samples. *p*-Cresol, which was positively correlated with the horsy, medicinal and sweaty saddle-like aroma descriptors, was the major contributor to the off-odor of SAB (Du et al., 2017). In addition, hexyl hexanoate, propyl hexanoate, ethyl caprate, 2-methylpropionic acid, ethyl heptanoate, and 1-octanol showed OAVs ≥1 in LZ–2016 but < 1 in GJ–1573. Notably, 3-methylindole (with animal, fecal-like odors) as well as 4-ethylguaiacol, 4-methyl guaiacol (with phenolic, clove-like odors), and 3-(methylthio)propanal (with a roasted potato-like odor) were detected with OAVs ≥1 only in the LZ–2016 base SAB. Among them, 3-(methylthio)propanal showed by far the highest OAVs. More importantly, as indicated by the data in Tables 1, 3-(methylthio)propanal was also found in the other 16 types of base SAB at concentrations varying between 183.6 and 1552.0 μg/L, and the OAVs ranged from 26 to 218. Thus, it was experimentally confirmed that 3-(methylthio)propanal is important and widespread in base SAB.

3.3. Aroma recombination and omission experiments

3.3.1. Reconstruction of the aroma profiles of the two SAB samples

To validate the quantitative results mentioned above, all the odorants with OAVs of ≥1 were combined in stock model solutions at their actual concentrations in LZ–2016 and in GJ–1573. Descriptive profile tests were performed separately for each aroma reconstruction in comparison to their corresponding original samples by rating the intensity of the eight aroma attributes as well as the overall similarity. As shown in Fig. 4A–B, the aroma profiles of the two tested SAB reconstitutions showed good similarity with those of the original samples. The overall aroma similarities between the recombination models and the original samples were judged to be 4.3 out of 5.0 points for LZ–2016 and 4.0 out of 5.0 points for GJ–1573. However, slight differences in aroma intensities compared with the original SAB samples remained, most notably differences in the roasted attributes for LZ–2016, and the roasted, green and mud-like attributes for GJ–1573.

3.3.2. Omission studies

To investigate the contributions of certain aroma compounds to the overall aroma profile of LZ–2016 base SAB, 13 omission models were prepared in which either a single component or a group of components with similar chemical structures and aroma descriptions were omitted (Table 5). Each of the omission models was evaluated by the triangle test in comparison with the complete model.

Not surprisingly, all the assessors reported that removing all esters (model 1) from the LZ–2016 base SAB resulted in a highly significant difference ($p < 0.001$) in the odor. Thus, esters were one of the key groups contributing to the fruity aroma of the base SAB. With respect to aromatic compounds, significant differences ($p < 0.001$) in the aroma characteristics of LZ–2016 base SAB were recognized by 7 out of 10 assessors (model 2). In addition, model mixtures without the sulfur compounds (model 6, which lacked 3-(methylthio)propanal and dimethyl trisulfide), and model 7, which lacked 3-methylindole were determined to be significantly different ($p < 0.01$) from the original

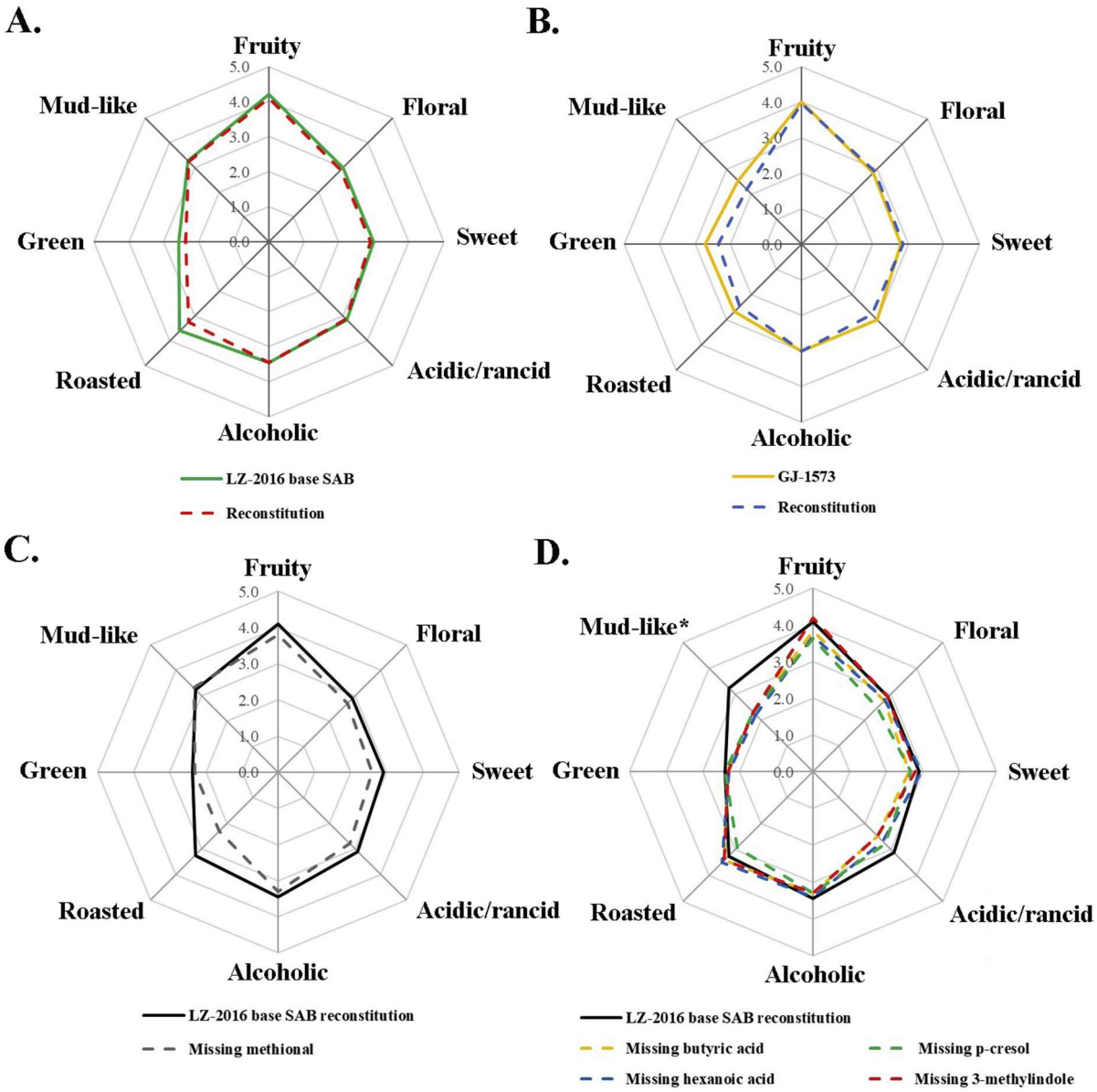


Fig. 4. Aroma profiles and reconstitution models of the two representatives SAB samples, (A) LZ–2016 base SAB and (B) GJ–1573, as well as the omission models of LZ–2016 base SAB (C and D).

Table 5
Omission experiments from LZ–2016 base SAB complete models.

NO.	Aroma	Omitted aroma compounds	Correct number in all	Significance[a]
1	Fruity	All esters	10/10	***
2	Floral	All aromatics	7/10	***
3	Acidic/rancid	All acids	4/10	*
3–1		Butyric acid	4/10	**
3–2		Hexanoic acid	5/10	*
4	Phenolic/horsy	All phenols	6/10	*
4–1		*p*-Cresol	7/10	*
5	Alcoholic	All alcohols	4/10	
6	Roasted/onion-like	All sulfur compounds	6/10	**
6–1		3-(Methylthio) propanal	5/10	*
6–2		Dimethyl trisulfide	8/10	**
7	Mud-like	3-Methylindole	6/10	**
8	Sweet	γ-Nonalactone	3/10	

[a] Significance: "*", "**", and "***" indicate significance at $p < 0.05$, 0.01, and 0.001, respectively.

solutions. Model 3 and model 4 were only slightly different ($p < 0.05$) from the base SAB. However, no significant differences were found when the alcohols (model 5) and γ-nonalactone (model 8) were removed. Therefore, the omission tests indicated that 3-(methylthio) propanal, dimethyl trisulfide, 3-methylindole, *p*-cresol, butyric acid, hexanoic acid, the esters, and the aromatics compounds listed in Table 5 were important and contributed substantially to the aroma profile of the LZ–2016 base SAB.

In recent decades, most studies on the aroma compounds of SAB suggested that the fruity, sweet, and floral odors were the most important odor characteristics; however, in this study, the roasted and mud-like attributes were chosen as the most relevant and differential odors for comparing the overall aroma profile of commercial SAB (GJ–1573) with that of base SAB (LZ–2016) for the first time. Therefore, descriptive analysis was used to further explore the contributions of certain aroma-active compounds to these two aroma characteristics. Models of five key aroma compounds, *p*-cresol (model 4–1), hexanoic acid (model 3–1), butyric acid (model 3–2), 3-(methylthio)propanal (model 6–1), and 3-methylindole (model 7), were

prepared by omitting the individual compound from the complete reconstitution. Among these compounds, hexanoic acid, butyric acid, and *p*-cresol were selected because of their importance indicated by the aforementioned omission tests; moreover, according to previous reports by Zhao and Du (Du et al., 2017; Zhao, Chen, et al., 2018), the removal of any of these three compounds from test model solution indicated they contributed substantially to the cellar and cheesy aromas of SAB, but little is known about their impacts on the roasted and mud-like odors. 3-(Methylthio)propanal and 3-methylindole were chosen because they were observed as important odorants in the base SAB for the first time by our group. More importantly, these two compounds have aroma descriptions that include roasted potato and mud-like; thus, it is necessary to check whether they can be used as marker compounds for these two odors. As shown in Fig. 4D, compared with the complete reconstitution of LZ–2016base SAB, the characteristics of the samples without *p*-cresol, hexanoic acid and butyric acid were significantly different from those of the base SAB, especially for their mud-like characteristics ($p < 0.05$). 3-Methylindole, which was previously identified by our group was contributing substantially to the mud-like odor characteristic, also showed a significant effect on the roasted and mud-like characteristics based on this omission study. When 3-(methylthio)propanal was removed (Fig. 4C), the intensities of the roasted aromas clearly decreased, however, the mud-like aromas slightly improved, which highlighted that 3-(methylthio)propanal in the base SAB not only was an important contributor to the roasted odor, consistent with the results of the SSB and RSB, but also is a typical potent odorant responsible for the mud-like aroma in the base SAB. Thus, these five compounds were the key odorants responsible for the roasted and mud-like aromas of the base SAB.

3.4. Interactions between the key aroma compounds responsible for the roasted and mud-like odors in the base SAB

To evaluate the interactions among the aforementioned five key compounds, the variations in the OAVs of the compounds before and after being mixed were compared and calculated as described in the "Analysis of the Interactions" section. In this experiment, the five tested compounds were classified into four combinations based on the resulting aromas. As shown in Table 6, the first mixture that was studied contained 3-(methylthio)propanal and butyric acid at concentrations consistent with those determined in the tested LZ–2016 base SAB sample. This mixture had to be diluted 136 times to reach the odor threshold, which means that the OAV of the mixture (OAV_{mix}) is 136. Interestingly, compared with the sum of the OAVs of the two individual compounds (Σ OAVs = 359), the multiplier of 3-(methylthio)propanal to butyric acid was 2.6, which means that a masking effect occurred between these two compounds. In addition, the effects of the three remaining mixtures were evaluated by a similar approach. Except for 3-(methylthio)propanal and *p*-cresol, which presented a weak additive effect, the other mixtures also showed masking effects with multipliers ranging from 3.3–5.3. Previous studies have demonstrated that the interactions among compounds are mainly influenced by the molecular size of the aroma compounds themselves, the type of functional groups, and the degree of their own volatility. For instance, mixtures of compounds (such as ethyl acetate and acetic acid) with dissimilar structures always show masking effects; however, this did not mean that all the mixtures with dissimilar structures presented a masking effect after mixing: a weak additive action was also evident between 2,4-hexadienal and phenyl acetaldehyde (Niu, Wang, et al., 2018). More importantly, until now, the reason for this remained unclear and was deemed to warrant further research.

In our study, although the chemical structures of 3-(methylthio) propanal and *p*-cresol are sufficiently different, they produced weak additive effects, which is different from the masking effects caused by 3-(methylthio)propanal and butyric acid, 3-(methylthio)propanal and hexanoic acid, and 3-(methylthio)propanal and 3-methylindole. Therefore, it is experimentally hypothesized that interactions between molecules might also be influenced by the hydrogen bonding, and more importantly, the hydrophobicity interactions. Regarding the chemical nature of each tested aroma compound, 3-methylindole (logP = 2.54) is more non-polar than hexanoic acid, *p*-cresol, and butyric acid (logP = 2.01, 1.91, and 1.0, respectively) (www.molinspiration.com). Basing our reasoning on a previous hypothesis of hydrophobicity interaction, the theorized intensity of the interactions between 3-(methylthio)propanal and the other four compounds might perform in a descending sequence of 3-methylindole, hexanoic acid, *p*-cresol, and butyric acid. Despite there being more non-polar structures, the complexity of the spatial conformation caused by the aromatic ring would significantly reduce the degree of interaction between them (Lorrain et al., 2013). Therefore, except for 3-(methylthio)propanal and *p*-cresol, which presented a weak additive effect, the other mixtures showed masking effects eventually.

Overall, the aforementioned results indicate that the interactions between compounds result in the roasted and mud-like aromas of base SAB, and these interactions are mainly additive and masking effects.

4. Conclusions

This study confirmed the scientific hypothesis that the roasted and mud-like aroma were both existed in the base SAB and the key aroma components responsible for aforementioned aromas were firstly determination, according to the cAEDA, multiple quantitation methods, and the calculation of OAVs. More importantly, based on omission analysis, hexanoic acid, butyric acid, *p*-cresol, 3-methylindole, and 3-(methylthio)propanal were confirmed to be the key odorants responsible for the mud-like and roasted odors in the base SAB. Additionally, the evaluation of the aforementioned five key odorants by the "addition" test showed that the roasted and mud-like aromas of base

Table 6
OAVs of mixtures of compounds in the ratios of their individual OAVs with their corresponding aroma descriptions.

No.	Mixture	spiking level (μg/L)	R[e]	SD[f]	Aroma description	Interaction relationship
1	3-(methylthio)propanal + butyric acid	[a]	2.6	0.23	cheesy, rancid, roasted potato, mud-like	masking effect
2	3-(methylthio)propanal + hexanoic acid	[b]	5.3	0.59	fruity, rancid, roasted	masking effect
3	3-(methylthio)propanal +3-methylindole	[c]	3.3	0.26	roasted potato, mud-like, fecal	masking effect
4	3-(methylthio)propanal + *p*-cresol	[d]	0.9	0.03	roasted, mud-like, animal-like, grain	additive action

[a] Mixture 1 = 369 μg/L 3-(methylthio)propanal +295,816 μg/L butyric acid.
[b] Mixture 2 = 369 μg/L 3-(methylthio)propanal +1,126,596 μg/L hexanoic acid.
[c] Mixture 3 = 369 μg/L 3-(methylthio)propanal +46 μg/L 3-methylindole.
[d] Mixture = 369 μg/L 3-(methylthio)propanal +3271 μg/L *p*-cresol.
[e] R = the sum of the individual OAVs of the compounds in the ratio of the OAVs of their mixtures.
[f] SD = The standard deviation of the R values.

SAB were related, and the odors were mainly attributed to additive or masking effects among the compounds for the first time. Those results provide basic scientific data for the quality control of Baijiu-making companies. However, further comprehensive investigations are necessary to identify the main precursors of the key aroma compounds responsible for the “roasted” and “mud-like” aromas in base SAB during the Baijiu brewing process.

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Notes

The authors declare no competing financial interest.

Declaration of Conflict of Interest

All the authors declare that they have no conflict of interest.

Appendix A. Supplementary data

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